

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: DEEP LEARNING

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO2: Analyze Machine Learning and Deep Learning models by evaluating neural network architectures, representation learning, activation functions, unsupervised learning techniques, and their real-world applications. (L4)

CO Weightage: 15%

Assessment Tool Weightage: 20%

QUESTIONS:

Total Marks: 20

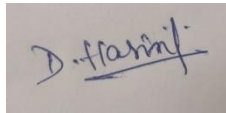
1. Differentiate between Machine Learning and Deep Learning with respect to architecture, data requirements, feature extraction and applications. Give two real-world applications of each.
2. Explain the concept of Representation Learning. Discuss how the width and depth of a neural network influence its learning capability, computational complexity and model performance.
3. Compare ReLU, Leaky ReLU (LReLU) and ELU (Exponential Linear Unit) activation functions. Explain their mathematical behavior, advantages, disadvantages and suitable use cases.
4. Explain the concept of unsupervised learning in neural networks. Describe the working principle of Restricted Boltzmann Machines (RBMs) and Autoencoders and compare their objectives and applications.

Code of Conduct:

I D.Hasini/192425147(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understanding	Basic understanding with some errors	Incorrect or very limited understanding
Content Coverage	25%	Covers all required topics comprehensively	Covers most topics with minor omissions	Covers some topics only	Very few topics covered
Technical Accuracy	20%	All explanations are technically correct and precise	Minor technical errors	Some incorrect or vague explanations	Major technical errors
Integration of Concepts	15%	Effectively connects and relates concepts logically	Some connections made with minor gaps	Limited connections between concepts	No clear relationship between concepts
Clarity & Presentation	15%	Well-structured, clear, and easy to understand	Mostly clear with minor issues	Somewhat unclear or poorly organized	Poorly structured and difficult to understand

Cog-AT3 - Class Test

1. Differentiate Between machine learning and Deep learning.

Concept understanding

Machine learning (ML) enables the systems to learn from data using algorithms, while deep learning is a subset of ML that uses multi-layered neural networks to automatically learn complex patterns.

Difference between ML & DL

Aspect	Machine learning	Deep learning
Architecture	Traditional algorithms. (Decision tree, SVM, KNN)	Multi-layer neural networks.
Data Requirement	Small to medium datasets	Large datasets
Feature Extraction	Manual	Automatic
Applications	Structured data	Images, Speech, Text.

Technical Accuracy

Machine learning requires manual feature engineering, whereas deep learning automatically extracts hierarchical features using hidden layers.

Integration of concepts

Deep learning extends machine learning by combining neural networks with representation learning to solve complex problems more effectively.

Applications:-

Machine Learning

- Email spam Detection.
- credit risk prediction in banking.
- customer churn prediction in telecom.
- sales forecasting.

Deep Learning

- Facial recognition systems.
- Autonomous robots.
- Object detection in surveillance systems.
- Video analysis and activity recognition.
- Recommendation systems.
like (Netflix, youtube, spotify).
- Drug discovery and genomics.

2. Representation learning, width and depth of neural networks

Concept understanding

Representation of learning enables neural networks to automatically learn useful features from raw data without manual feature extraction.

Content coverage

width:

- Number of neurons in a hidden layer.
- learns more features simultaneously
- Increases computational cost.

Depth:

- Number of hidden layers.
- learns hierarchical features.
- Improves accuracy for complex tasks but requires more computational.

Technical Accuracy

Increasing width improves feature learning, while increasing depth enables the network to capture higher level representations.

Integration of concepts.

- Representation learning automatically extracts meaningful features.
- width increases the number of features learned simultaneously
- Depth enables hierarchical feature learning from simple to complex patterns.
- Together, They improve model accuracy, generalization and performance in real-world applications.

Applications: of Representation learning

- Image classification
- facial recognition
- Anomaly detection in sensor data.
- Data compression.

Applications of wider and deeper neural networks

- video surveillance.
- voice assistants.
- predictive maintenance.
- Handwriting Recognition.

3) ReLU, leaky ReLU, and ELU

Concept understanding

Activation functions introduce non-linearity, enabling neural networks to learn complex relationships.

Content coverage

Activation function: ReLU

- Advantages:- fast, simple, avoids vanishing gradient
- Disadvantages:- Dying ReLU problem
- use cases:- CNN, Image Recognition.

Leaky ReLU

- Advantages:- prevents dying ReLU
- Disadvantages:- Requires α parameter
- use cases:- Deep Networks, GANs.

ELU

- Advantages:- faster convergence, zero mean outputs
- Disadvantages:- computationally expensive.
- Use cases:- Medical Imaging, NLP.

Technical Accuracy

- ReLU: $f(x) = \max(0, x)$
- leaky ReLU: small negative slope for negative inputs.
- ELU: exponential output for negative inputs.

Integration of concepts:

leaky ReLU and ELU improve upon ReLU by maintaining gradient flow, leading to better learning in deep networks.

Applications

ReLU:

- convolutional neural networks (CNN)
- face recognition processing
- Autonomous vehicles.

leaky ReLU

- Generative Adversarial networks
- Speech recognition
- financial prediction models.

ELU

- Time-series forecasting
- Healthcare Analysis.

4. unsupervised learning, RBMs and Autoencoders.

concept understanding:-

unsupervised learning trains neural networks using unlabeled data to discover hidden patterns and meaningful representations.

Restricted Boltzman machine (RBM)

- visible & hidden layers.
- learns probability distribution of data.
- uses contrastive Divergence.

Auto encoder.

- encoder → Bottleneck → Decoder.
- learns compressed representations.
- Reconstructs input by minimizing reconstruction error.

Technical Accuracy:

RBM

- probabilistic model.
- learns hidden probabilities.
- Recommendation system

Autoencoder.

- Neural network model.
- learns compressed features.
- feature extraction and reconstruction.

Integration of concepts:

Both RBMs and Autoencoders perform unsupervised representation learning, but RBMs focus on learning data distributions, whereas Autoencoders focus on learning compact feature representations for reconstruction and analysis.

Applications:-

RBM

- Topic modeling
- Deep Belief networks (DBNs)
- Collaborative filtering
- Feature extraction.
- Music recommendation.

Autoencoders:

- Dimensionality Reduction.
- Customer Segmentation.
- Network Intrusion Detection
- Missing Data reconstruction.